

AI-BASED PREDICTIVE MODELS FOR IDENTIFYING GAZELLES

Kiril Anguelov

*Technical University,
Sofia, Bulgaria,*

ABSTRACT

This article examines the role of fast-growing enterprises (“gazelles”) in economic development and the challenges associated with their early identification. Classical approaches such as financial ratios, multi-criteria methods and regression models are analyzed, as well as their limitations in capturing complex dependencies. Based on a review of the current literature, a conceptual model is proposed that uses machine learning algorithms – Random Forest, XGBoost and neural networks – and compares them with basic models such as logistic regression. Emphasis is placed on the evaluation metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC) and the application of Explainable AI techniques (SHAP), which allow for interpretation of the results. The study highlights the potential of AI to increase the accuracy and reliability of forecasts, providing practical value for investors, incubators and government policies.

Key words: gazelles, predictive models, machine learning, high-growth firms

INTRODUCTION

Growth and innovation, inextricably linked to the development of the modern economy, are fundamental to the emergence and development of the so-called gazelle companies. It is categorically clear that these are companies that in number are a relatively small part of the overall business, but at the same time it is equally clear that they have a huge, disproportionately large contribution to the emergence of new jobs, opportunities for improving efficiency and competitiveness, as well as their important role in the implementation of new technologies. From this point of view, the place and role of gazelles comes quite logically, which usually enjoy strong interest from investors, business incubators and even governmental and European policy, seeking to stimulate sustainable economic development.

Among the main challenges faced by both practitioners and researchers is the difficulty of identifying at an early stage companies that have the potential to become gazelles in the coming years. The difficulty can be traced to the complexity of the business, as well as the different intensity of numerous environmental factors that influence the company. The so-called business dynamics, in turn, are subject to multiple, interrelated elements – from quick strategic decisions of manage-

ment and its managerial capacity and innovative spirit, to the possibility of access to financing, state or specific support policies, and many others.

In this context, the new opportunities that AI has the capacity to offer are outlined, in order to approach this problem in a new way - the early identification of companies with the potential of gazelles. The ability of AI to process and analyze huge datasets, discovering patterns and relationships in them that may not be so obvious to the human eye, can also be used to identify potential gazelles. AI-based predictive models can facilitate the analysis of financial and operational indicators. At the same time, they can add various sources of information, which can add additional reliability and validity to forecasts for future development.

The aim of this study is to develop and present a conceptual model, based on artificial intelligence, for early identification of fast-growing enterprises (“gazelles”), by combining classic financial indicators with advanced machine learning algorithms and methods for interpretability of results.

LITERATURE REVIEW

In the economic literature, the term “high-growth enterprises” (HGEs) refers to those firms that, in a relatively short period of time, achieve significant growth in key indicators. The joint definition of Eurostat and OECD defines HGE as an enterprise with at least ten employees in the base year, which achieves an average annual growth rate of at least 20% in turnover or employment for three consecutive years. A subset of these enterprises are the so-called “gazelles” – firms with the same growth dynamics, but up to five years old (Eurostat, 2024a; Eurostat, 2024b; OECD, 2010). National statistical offices in a number of countries apply this framework in their own measurements. For example, in the United Kingdom, the OECD definition is used in the analysis of the innovation ecosystem (UK Government, 2022), and in Ireland, the Central Statistics Office publishes detailed methodological notes, consistent with European standards (CSO Ireland, 2023).

Gazelle growth theory shows that organic growth and acquisitions are interrelated, with the lower cost of organic growth and the availability of financial support encouraging gazelle companies to expand their development through acquisitions (Kaya & Persson, 2019). An analysis of firms in Sweden (1990–2016) shows that

despite the fact that most high-growth firms („gazelles“) are small, the probability of high growth is lowest for small old firms, despite their large numerical presence (Coad & Karlsson, 2022). A study of an accelerator in Colombia (González-Urbe & Reyes, 2021) shows that high growth in developing economies is limited primarily by a deficit in corporate capacities, and alleviating them through training and support can unleash the potential of innovative entrepreneurs, but does not turn weak ideas into high-growth companies. A systematic review of 66 articles shows that the early success of “unicorns” and “gazelles” depends on factors such as the rapid scalability of business models and the key role of early investors, whose support accelerates survival but also poses a risk in case of withdrawal (Cristofaro, M., Abatecola, G., Giannetti, F., Zannoni, A. (2024).

TRADITIONAL METHODS FOR GAZELLES IDENTIFICATION

Gazelle research is associated with a variety of analytical approaches. The basis of classical methods are financial ratios – return on equity (ROE), return on investment (ROI), turnover growth and other indicators, which traditionally serve to assess the efficiency and probability of sustainable growth. Empirical results confirm that it is these indicators that often predict the future status of HGF (Srhoj, 2022).

Another approach includes multi-criteria decision-making methods (MCDM), such as TOPSIS, MOORA/MULTIMOORA, WASPAS and their extensions. They allow for the evaluation and ranking of enterprises according to multiple criteria, including qualitative and quantitative indicators, and are applied both in supply chain management and in strategic analysis of companies (Behzadian et al., 2012; Taherdoost & Madanchian (2023) etc.).

A third area of research in the literature is regression models, often based on Gibrat’s law of proportional growth. Logit and probit models, panel and quantile regressions are used to examine the relationships between firm size, age and growth. Reviews by Santarelli & Thurik (1999) show that while regression methods provide good interpretability, they struggle to capture the nonlinearities and complex factors that drive rapid growth.

ARTIFICIAL INTELLIGENCE IN BUSINESS FORECASTING

With the development of artificial intelligence over the past decade, increasing attention has been paid to the possibility of applying machine learning to forecast company growth. Algorithms such as Random Forest, Gradient Boosting, and Support Vector Machines show

significant advantages when working with large and heterogeneous data sets. Comparative studies demonstrate that these models often outperform classical regressions in accuracy and robustness. Specifically for high-growth companies, studies are already emerging that use XGBoost, Random Forest, and logistic regression to classify companies as “gazelles” and achieve promising results. In addition to traditional machine learning, deep learning techniques are gradually gaining ground in the literature. Neural networks, including recurrent and convolutional architectures, have found applications in financial time series forecasting and in ensemble models for business forecasting. A key aspect of AI application in economics is the need for transparency. Explainable AI concepts, such as SHAP and LIME, allow for local and global explanation of predictions, which is particularly important in investment and policy decision-making.

In the contemporary literature on firm growth prediction, particular attention is paid to machine learning algorithms that can handle large, heterogeneous, and often incomplete data sets. Among them, Random Forest (RF) is one of the most widely used. This model is an ensemble of multiple decision trees, each of which is trained on different subsets of data and features. The main advantage of RF is its resistance to overfitting and its ability to capture nonlinear dependencies and complex interactions between variables. In studies on high-growth firms, RF is often used as a basic ensemble method, which offers a good balance between accuracy and interpretability by inferring the importance of individual features.

Another leading approach is XGBoost and the wider family of Gradient Boosting Machines (GBM). Unlike RF, where trees are built in parallel, in gradient boosting they are constructed sequentially, with each new tree correcting the errors of the previous ones. This process leads to high predictive accuracy and allows modeling of complex nonlinear relationships. XGBoost, in particular, is optimized for speed and efficiency and has become a standard in a number of application tasks, including classifying companies according to their growth. Research shows that GBM and its variants often outperform RF in tasks with high class imbalance, such as the situation in forecasting gazelles.

Neural networks are another important tool that is gaining attention in business forecasting. They can handle complex, nonlinear, and high-dimensional data by using layers of artificial neurons to extract representations. In the context of company growth, neural networks can be applied to time series (e.g., turnover or employment over time) as well as to combined data including financial and non-financial indicators. Although

they require larger computational resources and careful tuning, these models have the potential to identify hidden patterns that remain inaccessible to classical statistical methods.

To assess the real advantage of the above AI algorithms, they are usually compared to baseline models, such as logistic regression. Logistic regression has a long tradition in the analysis of firm growth and is preferred for its simplicity and interpretability. It allows for clear identification of the relationships between predictors and the probability of a firm being classified as a “gazelle”. However, logistic regression has difficulty capturing complex nonlinearities and interactions, which makes it less competitive compared to modern AI methods. That is why it is often used as a reference model against which the advantages of RF, GBM and neural networks are evaluated.

METRICS FOR EVALUATION

The performance of AI models for predicting fast-growing companies should be measured by a combination of metrics that reflect different aspects of the classification task. The most basic metric is Accuracy, which measures the proportion of correctly classified observations out of the total. Although widely used, this metric can be misleading in conditions of imbalance between classes, as is the case with “gazelles”, which represent a small part of the population of companies.

Therefore, attention is also paid to Precision and Recall. Precision shows what proportion of companies classified as “gazelles” really fall into this category, while sensitivity measures how many of the actual “gazelles” are correctly identified by the model. These two metrics are often combined into an F1-score, which represents a harmonic mean between them and offers a balanced assessment in conditions of imbalance.

Another key tool for comparing models is ROC-AUC (area under the receiver operating characteristic curve). The ROC curve depicts the relationship between sensitivity and specificity at different classification thresholds, and the AUC summarizes this information in a single value. A high AUC value indicates a good ability of the model to distinguish gazelles from other firms.

In addition to assessing overall performance, it is particularly important to understand which factors have the greatest impact on predictions. In this context, Feature Importance indicators are used, which indicate the relative importance of different variables. In modern research, Explainable AI methods, such as SHAP values, are increasingly being applied, which provide both global and local explanations for the role of individual indicators. SHAP can identify, for example, whether turnover dynamics, the level of innovation ac-

tivity or capital structure are key to the probability of a firm being classified as a gazelle. Thus, the evaluation of models is not limited to numerical accuracy indicators, but also provides valuable managerial and political information.

CONCLUSION

The analysis shows that the application of artificial intelligence in the forecasts of fast-growing companies opens up new horizons for economic research and practice. While traditional methods – financial ratios, multi-criteria estimates and regression models – remain a valuable tool for initial diagnostics, they hardly capture the complex and dynamic nature of “gazelles”. Algorithms such as Random Forest, XGBoost and neural networks demonstrate higher predictive power and the ability to process a variety of data, including non-financial indicators. The role of Explainable AI is particularly significant, since methods such as SHAP can explain the key factors behind the predictions and provide confidence in the use of these models in practice.

It can certainly be considered that AI-based approaches contribute to faster and more accurate business forecasts, suitable for making competent strategic management decisions. At the same time, they provide valuable information and a comprehensive toolkit not only for managers, but also for investors, business incubators, and in a broader context - opportunities for targeted support from government institutions and international organizations.

REFERENCES:

1. Behzadian, S., Otaghsara, K., Yazdani, M., Ignatius, J. (2012). A state-of the-art survey of TOPSIS applications, *Expert Systems with Applications*, Volume 39, Issue 17, Pages 13051-13069, ISSN 0957-4174, <https://doi.org/10.1016/j.eswa.2012.05.056>.
2. Coad, A., Karlsson, J. (2022). A field guide for gazelle hunters: Small, old firms are unlikely to become high-growth firms, *Journal of Business Venturing Insights*, Volume 17, e00286, ISSN 2352-6734, <https://doi.org/10.1016/j.jbvi.2021.e00286>.
3. Cristofaro, M., Abatecola, G., Giannetti, F., Zannoni, A. (2024). The survival of the fastest: Unveiling the determinants of Unicorns and Gazelles' early success, *Scandinavian Journal of Management*, Volume 40, Issue 2, 101335, ISSN 0956-5221, <https://doi.org/10.1016/j.scaman.2024.101335>.
4. González-Urbe, J., Reyes, S. (2021). Identifying and boosting “Gazelles”: Evidence from business accelerators, *Journal of Financial Economics*, Volume 139, Issue 1, Pages 260-287, ISSN 0304-405X, <https://doi.org/10.1016/j.jfineco.2020.07.012>.
5. Kaya, M. C., Persson, L. (2019). A theory of gazelle

growth: Competition, venture capital finance and policy, The North American Journal of Economics and Finance, Volume 50, 101019, ISSN 1062-9408, <https://doi.org/10.1016/j.najef.2019.101019>.

6. Santarelli, E., & Thurik, A. R. (1999). Gibrat's Law: An overview of the empirical literature. In Z. J. Acs & D. B. Audretsch (Eds.), *Industrial evolution: Entrepreneurship and the growth of firms* (pp. 183–226). Cambridge University Press. <https://doi.org/10.1017/CBO9780511758683.009>.

7. Srhoj, S. (2022). Can we predict high growth firms with the financial ratios?, *Financial Internet Quarterly*, ISSN 2719-3454, Sciendo, Warsaw, Vol. 18, Iss. 1, pp. 66-73, <https://doi.org/10.2478/fiqf-2022-0006>.

8. Taherdoost, H., Madanchian, M. (2023). A Com-

prehensive Survey and Literature Review on TOPSIS (November 07, 2023). Available at SSRN: <https://ssrn.com/abstract=5262139> or <http://dx.doi.org/10.2139/ssrn.5262139>.

9. Central Statistic Office Ireland (2023), <https://www.cso.ie/en/methods/surveybackgroundnotes/highgrowthenterprises>

10. Eurostat, High-growth enterprise definition, <https://ec.europa.eu/eurostat>

11. OECD (2010), High-Growth Enterprises: What Governments Can Do to Make a Difference, OECD Studies on SMEs and Entrepreneurship, OECD Publishing. <http://dx.doi.org/10.1787/9789264048782-e>

12. United Kingdom Government (2024), <https://www.gov.uk/government/publications/>